PYTHON CONVERSATIONAL LOOP

Rule-Based Chatbot — Professional Task Submission

Student Name	D.Chandrika
Registration Number	99230040519
Technology	Python 3
Project Type	Rule-Based Conversational Chatbot

1. Objective

Build a simple rule-based conversational chatbot in Python. The chatbot continuously interacts with the user, recognizes different messages, provides appropriate responses, and allows the user to exit the conversation.

2. Design Approach

Input: The chatbot receives a message from the user. **Process:** The message is normalized and checked against predefined rules. **Respond:** A suitable response is selected. **Repeat:** The process continues inside a loop. **Exit:** The loop ends when the user enters *bye* or *exit*.

3. Complete Python Implementation

```
def chatbot():
    print("====================================")
    print("      Welcome to Smart Chatbot")
    print("====================================")

    name = input("What is your name? ").strip()

    print(f"\nNice to meet you, {name}!")
    print("You can ask me about Python, AI, or college.")
    print("Type 'bye' or 'exit' to end the conversation.\n")

    while True:
        user_input = input(f"{name}: ").lower().strip()

        # Exit condition
        if user_input == "bye" or user_input == "exit":
            print(f"Chatbot: Goodbye, {name}! Have a great day.")
            break

        # Greeting recognition
        elif "hello" in user_input or "hi" in user_input:
            print(f"Chatbot: Hello {name}! How can I help you?")

        # Python topic
        elif "python" in user_input:
            print("Chatbot: Python is a high-level programming language.")
            print("It is widely used in web development, automation,")
            print("data science, and AI.")

        # Artificial Intelligence topic
        elif "ai" in user_input or "artificial intelligence" in user_input:
            print("Chatbot: Artificial Intelligence enables computers to")
```

```
            print("perform tasks that normally require human intelligence,")
            print("such as learning and decision-making.")

        # College / placement topic
        elif "college" in user_input or "placement" in user_input:
            print("Chatbot: College placements help students connect")
            print("with companies and explore career opportunities.")

        # Help option
        elif "help" in user_input:
            print("Chatbot: You can ask about Python, AI, college,")
            print("or placements.")

        # Unknown input
        else:
            print("Chatbot: I'm still learning. Please try asking")
            print("about Python, AI, college, or placements.")


if __name__ == "__main__":
    chatbot()
```

4. Requirement Mapping

PDF Requirement	Implementation in Solution
Ask user's name	name = input(...)
Use while True	Continuous conversation loop
Use input()	Reads every user message
Use .lower().strip()	Normalizes each message
At least two greetings	hello and hi
At least three topics	Python, AI, college/placement
Use if / elif / else	Rule-based decision structure
Use break	Stops on bye or exit
Use user's name	Greeting and goodbye responses
Useful unknown response	Final else block
Use in for flexibility	Recognizes words in longer messages

5. Sample Test Run

```
==================================
      Welcome to Smart Chatbot
==================================
What is your name? Chandrika

Nice to meet you, Chandrika!

Chandrika: HELLO
Chatbot: Hello Chandrika! How can I help you?

Chandrika: tell me about python
Chatbot: Python is a high-level programming language.
It is widely used in web development, automation, data science, and AI.

Chandrika: what is AI
Chatbot: Artificial Intelligence enables computers to perform
tasks that normally require human intelligence.

Chandrika: hello bot
Chatbot: Hello Chandrika! How can I help you?

Chandrika: something random
Chatbot: I'm still learning. Please try asking about Python, AI,
college, or placements.

Chandrika: bye
Chatbot: Goodbye, Chandrika! Have a great day.
```

6. Technical Explanation

The chatbot follows a simple rule-based architecture. User input is first normalized using **lower()** and **strip()**. The program then evaluates the message through an **if / elif / else** decision chain. The **in** operator allows partial matching, which improves flexibility for inputs such as “hello bot”. The **while True** loop maintains the conversation, while **break** provides a controlled exit mechanism.

7. Viva / Demonstration Questions

Question	Professional Answer
What type of chatbot is this?	It is a rule-based chatbot that responds according to predefined conditions.
Why is while True used?	It keeps the conversation running until an explicit exit condition is reached.
Why use lower()?	It makes matching case-insensitive, so HELLO and hello can be handled similarly.
Why use strip()?	It removes unwanted spaces around the user's input.
Why use the in operator?	It lets the chatbot recognize keywords within longer messages.
What does break do?	It immediately terminates the conversation loop.
How does the chatbot handle unknown input?	The final else block provides a useful fallback response.
Can this chatbot learn automatically?	No. This implementation follows predefined rules and does not train on new data.

Execution Flow: INPUT → PROCESS → RESPOND → REPEAT → EXIT